

Alvaro Perdomo Gutierrez

Senior Full Stack Engineer

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with **12+** years across fintech, healthcare, and e-commerce, building scalable microservices and cloud-native applications. Expert in Java Spring Boot, Node.js, and React with hands-on experience in microservices architecture, REST/GraphQL APIs, and cloud platforms including Azure and AWS. Proven track record delivering end-to-end modules, optimizing system performance, and integrating AI capabilities into production systems.

SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL, HTML, CSS
Backend: Spring Boot, Node.js, Express, NestJS, FastAPI, Kafka, RabbitMQ, gRPC
Frontend: React, Next.js, Redux, Tailwind CSS, Material-UI, Webpack, Vite
Database: PostgreSQL, MongoDB, Cosmos DB, Redis, Elasticsearch, TcaplusDB
Testing: Jest, Pytest, Playwright, Cypress, JUnit, Selenium
Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, GitHub Actions, Jenkins, Terraform, CloudFormation, Azure DevOps
Others: RESTful API, GraphQL, Microservices, Event-Driven Architecture, Git, Jira, Agile, Grafana, Prometheus, Datadog, OpenTelemetry

EXPERIENCE

Leanware May 2024–Present
Senior Software Engineer

- Spearheaded the design and development of the AI-powered fitness platform backend using **Java** and **Spring Boot**, architecting microservices that processed real-time user interactions and ML predictions while maintaining sub-500ms response times across on-demand feature deployments.
- Engineered full-stack features for an AI fitness application using **React** on the frontend and **Node.js** backends, collaborating with the data team to integrate machine learning models that personalized workout recommendations, increasing user engagement by 18%.
- Architected a scalable event-driven system using **Kafka** to handle asynchronous processing of fitness analytics and user behavior telemetry, reducing API response latency by 35% and enabling real-time notifications across 50K+ concurrent users.
- Implemented comprehensive REST and **GraphQL** API layers using **Spring Boot** and **Node.js**, establishing clear contracts between frontend and backend services while supporting multiple client types including mobile and web applications.

- Deployed containerized microservices to **AWS** and **Kubernetes** clusters using infrastructure-as-code principles with **Terraform**, managing database migrations and ensuring zero-downtime deployments across staging and production environments.
- Integrated OpenAI API and LLM-based features into the platform backend, building prompt orchestration workflows that powered AI-driven fitness coaching and personalized meal planning recommendations without compromising data privacy.
- Mentored two junior engineers on microservices patterns and cloud-native development, conducting architecture reviews and establishing coding standards that reduced production incidents by 22% over six months.
- Optimized database queries and implemented **Redis** caching for frequently accessed user profiles and workout templates, cutting database load by 40% during peak traffic hours and improving page load time from 1.2s to 340ms.
- Established observability infrastructure by configuring **Prometheus** and **Grafana** dashboards to monitor service health, latency, and error rates across the fitness platform, enabling faster incident resolution and reducing mean time to recovery by 50%.
- Drove CI/CD pipeline improvements using **GitHub Actions**, parallelizing test execution and automating Docker image builds, which cut deployment time from 18 minutes to 4 minutes and improved team deployment confidence.

ThredUp

Mar 2022–May 2024

Senior Software Engineer

- Led the migration of the fashion resale platform's ML data infrastructure to **Databricks**, reducing model training cycles from 2 quarters to 4 weeks while enabling rapid experimentation with garment-recognition algorithms that processed 100K+ daily listings.
- Owned end-to-end development of the AI Shopping module using **React** for the frontend and **Node.js** microservices for backend inference, integrating image-search and style-chat features that launched in production within 6 weeks and drove a **15%** increase in seller conversion.
- Built scalable REST APIs using **Java** and **Spring Boot** to surface real-time inventory pricing predictions and automated photo-generation results to mobile and web clients, handling 172M+ items across 55K brands with sub-300ms latency requirements.
- Architected event-driven data pipeline using **Kafka** to stream product tagging events and visual-recognition model outputs to downstream analytics systems, processing 275K+ images weekly and reducing manual quality-assurance overhead by 30%.
- Implemented **Redis** caching layer for frequently accessed seller profiles and product metadata, reducing **PostgreSQL** query load by 45% and enabling real-time personalization recommendations that increased average order value by 2.8% on iOS.
- Deployed containerized Python and Node services to **AWS** using **Docker** and **Kubernetes**, automating database schema migrations with zero downtime and establishing runbooks for on-call incident response that halved production alert volume.

- Collaborated with the ML team to operationalize dropout-risk prediction and demand-forecasting models, building **GraphQL** resolvers that provided upstream systems with confidence scores and flagged anomalies for human review.

Spring Health

Nov 2020–Mar 2022

Software Engineer

- Played a key role in scaling the precision-matching ML platform backend from serving 300K to 2M+ members across a HIPAA-compliant infrastructure, refactoring matching algorithms in **Java** to handle 6x growth without degrading recommendation accuracy.
- Engineered the global Care Navigation platform frontend using **React** and backend services in **Node.js**, enabling dependent eligibility matching and jurisdiction-aware clinical routing for Fortune 500 clients including PepsiCo and Instacart.
- Built resilient microservices using **Spring Boot** that integrated with legacy claim systems and EHR data sources via SOAP and REST interfaces, maintaining strict HIPAA audit logging and encryption requirements for healthcare compliance.
- Developed operational ML services for dropout-risk prediction and demand forecasting, training models that identified at-risk members with 87% precision, enabling proactive outreach that reduced member churn by 5% annually.
- Implemented **PostgreSQL** schema redesigns to support complex matching logic across clinical condition hierarchies and insurance plan types, optimizing join performance and reducing matching latency from 8s to 1.2s per member query.
- Deployed microservices to **AWS** using **Docker** containers and **Kubernetes** orchestration, setting up automated scaling policies that adapted to seasonal demand spikes in benefits enrollment periods without service degradation.

Tencent Games

Nov 2014–Nov 2020

Senior Software Engineer

- Contributed to the backend architecture of Honor of Kings, a mobile MOBA that scaled to 100M daily active users, implementing account services and matchmaking systems in **Java** and **C++** that handled concurrent session management across millions of players.
- Built game-server infrastructure components using **Node.js** to support cross-platform progression synchronization for Arena of Valor's global launch, enabling players to seamlessly switch between mobile and console while maintaining character state consistency.
- Designed and maintained database schemas in **TcaplusDB** (Tencent's proprietary distributed database) handling 40M+ daily player sessions, implementing caching strategies with **Redis** that reduced database load during peak hours and improved login speed by 28%.
- Engineered the anti-cheat system integration with kernel-level ACE module, working with the security team to detect and prevent unauthorized client modifications in multiplayer battles, reducing fraudulent accounts by 19%.

- Developed API contracts using gRPC and REST for communication between game-client, game-server, and backend services, ensuring low-latency message delivery critical for real-time competitive gameplay across APAC regions.
- Collaborated with DevOps teams to containerize legacy game-server code using **Docker**, deploying services to **GCP** infrastructure and establishing monitoring via **Prometheus**, reducing deployment friction and enabling weekly release cadences.

EDUCATION

The Hong Kong Polytechnic University
Bachelor of Science in Computer Science

Sep 2010–Nov 2014